

Inconsistencies in the role of the tongue root in phonological contrasts

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Central questions

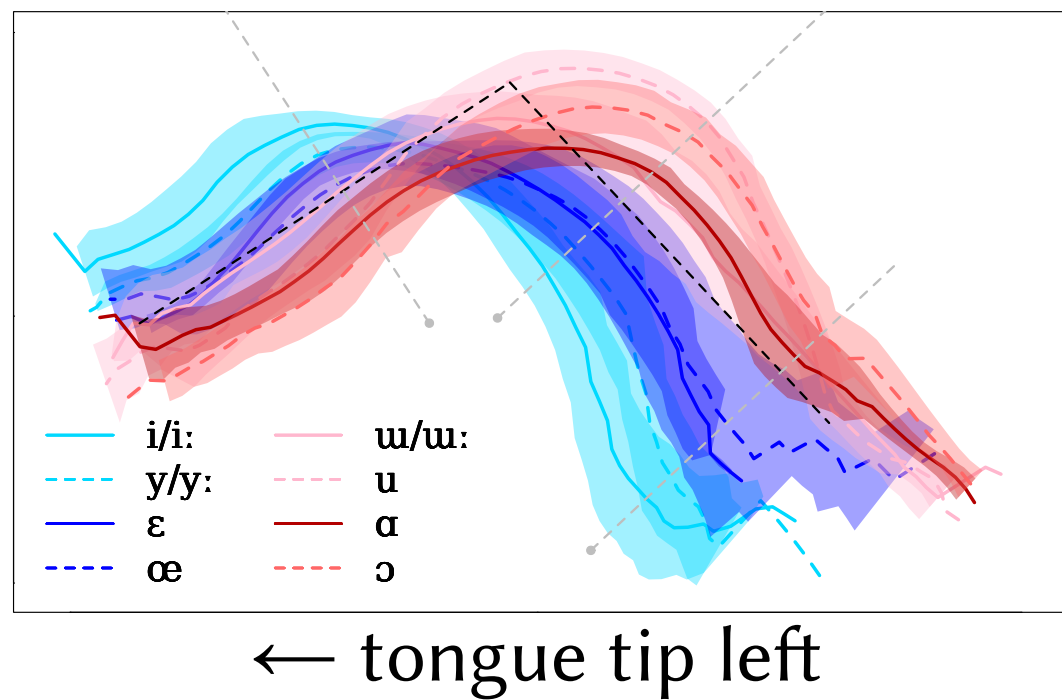
- Is the role of the tongue root (TR) quantifiably consistent with previous studies?

Approach

- Develop quantifiable measure of contribution of various areas of tongue
- Test on various contrasts reported to use TR

Dolgan

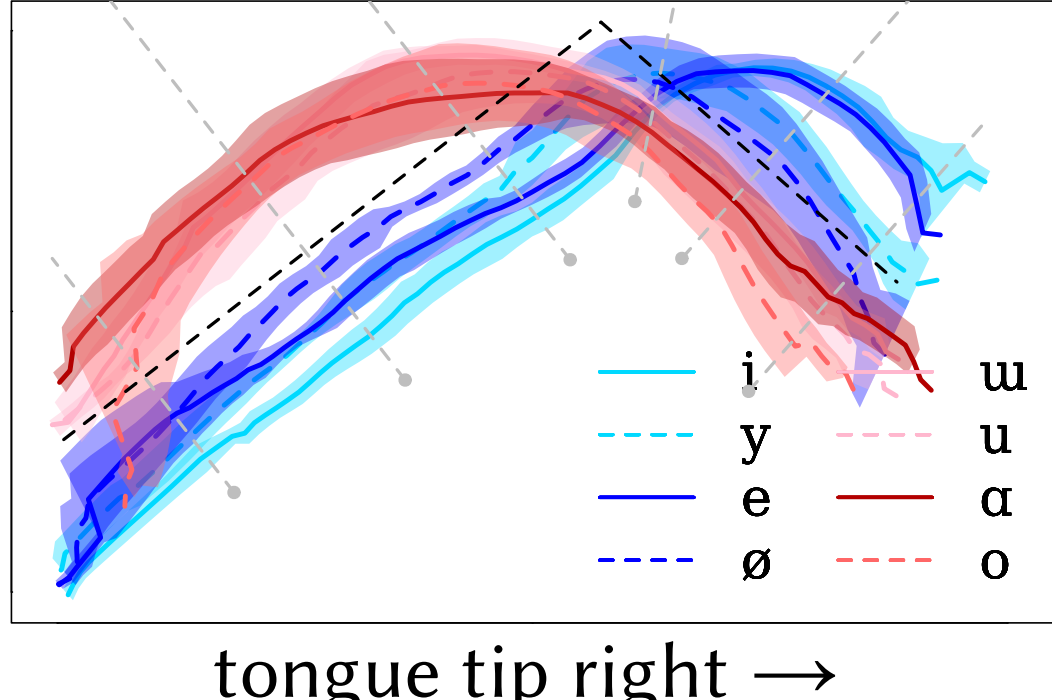
V height: TR↔, TB↑
V backness: TR↔, TB↔
(Banach et al., to appear)



	pair	middle	TB	PMC	upper	lower
height	i-e	0.00052	1.00000	0.28013	0.09608	
	y-ø	0.02035	1.00000	1.00000	0.19771	
	u-ɑ	0.00555	0.00049	1.00000	1.00000	
	u-ɔ	0.16815	0.01150	0.02140	1.00000	
	i-u	1.6e-07	0.20473	3.5e-08	2.4e-05	
	y-u	1.3e-07	4.3e-13	3.8e-13	1.2e-07	
	ε-ɑ	5.0e-06	1.00000	0.11247	1.8e-09	
	ε-ɔ	2.5e-06	4.2e-10	2.7e-08	0.00252	
	i-y	0.00732	0.00584	0.12560	1.00000	
	ε-ø	1.00000	1.00000	1.00000	1.00000	
	ɔ-ɑ	0.27664	1.3e-06	4.3e-07	0.03711	
	u-u	0.07095	1.4e-06	1.4e-05	1.00000	

Kyrgyz

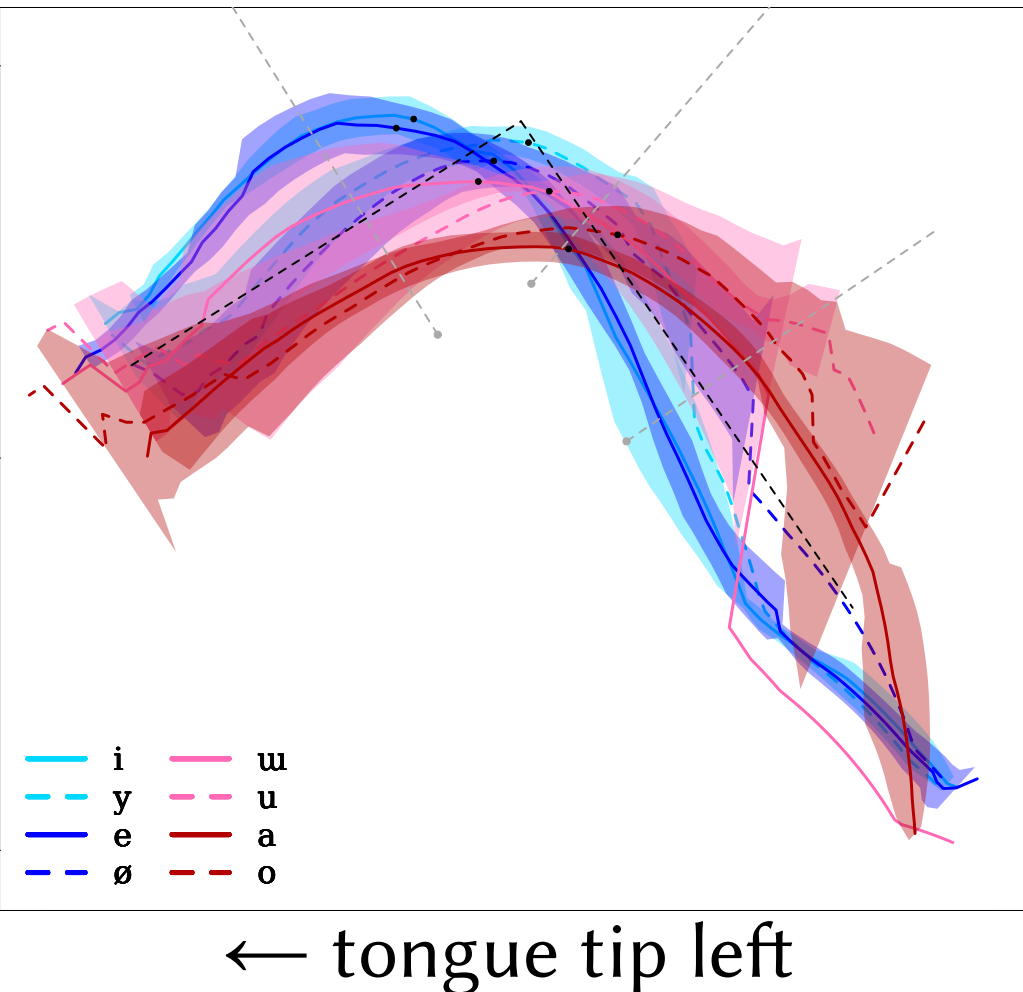
V height: TB↑
V backness: TB↔, TR↔
(Washington, 2016, 2019)



	pair	middle	TB	front	upper	TR mid	lower
height	i-e	1.00000	1.00000	0.5289	0.03939	0.77780	
	y-ø	1.00000	1.00000	0.0194	0.01146	1.00000	
	u-ɑ	1.00000	1.00000	1.0000	1.00000	0.55062	
	u-ɔ	1.00000	0.46607	1.0000	1.00000	1.00000	
	i-u	1.1e-05	8.9e-05	8.3e-08	1.6e-08	0.00038	
	y-u	0.01276	0.00106	0.0123	0.00060	6.0e-05	
	ε-ɑ	9.4e-07	1.2e-05	0.0069	4.7e-05	0.00744	
	ε-ɔ	0.15213	0.00726	0.0907	0.00044	1.00000	
	i-y	0.02002	0.00339	0.0299	0.04861	1.00000	
	ε-ø	0.19257	0.00381	3.0e-05	0.00789	1.00000	
	ε-ɔ	1.00000	0.41320	1.0000	1.00000	1.00000	
	u-u	1.00000	1.00000	1.0000	1.00000	1.00000	

Qumuq

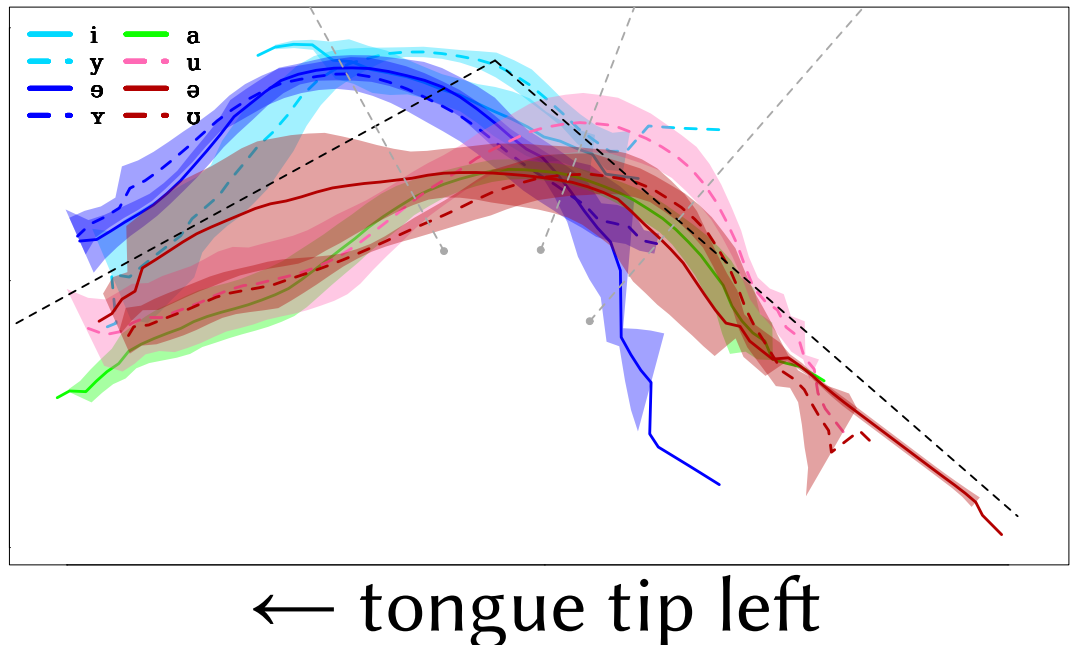
V height: TB↑
V backness: TB↔
(Washington et al., 2020)



	pair	middle	upper	middle
height	i-e	1.00000	1.00000	1.0000
	y-ø	1.00000	0.91688	ND
	u-ɑ	0.37080	0.03259	1.0000
	u-ɔ	1.00000	0.15022	1.0000
	i-u	0.89866	1.00000	1.0000
	y-u	0.05915	1.0000	ND
	ε-ɑ	3.8e-06	1.00000	9e-06
	ε-ɔ	0.02417	0.04722	1.0000
	ø-ɑ	0.00041	0.25790	ND
	i-y	0.00041	0.25790	ND
	ε-ø	0.00060	0.26185	0.5554
	ɑ-ɔ	1.00000	1.00000	1.0000
	u-u	0.93427	1.00000	1.0000

Tatar

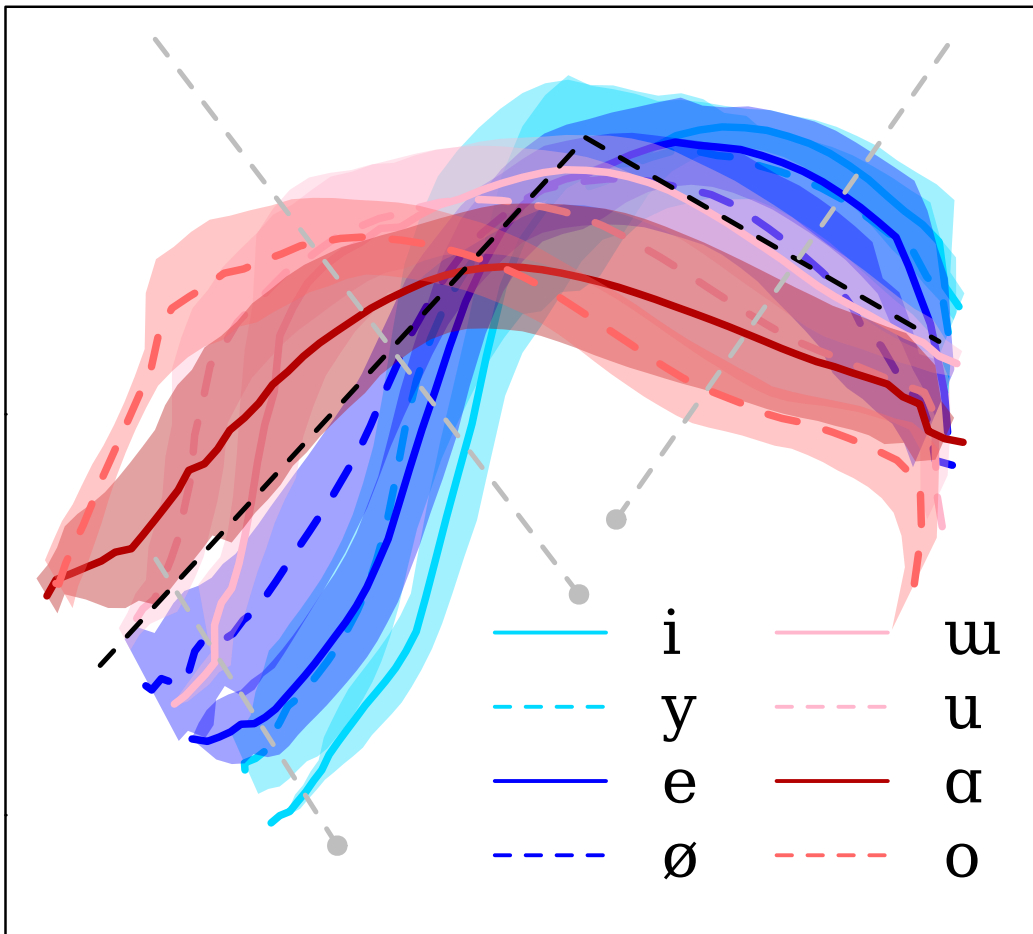
V height: TB↑
V backness: TB↔, TR↔
(Washington et al., 2020)



	pair	middle	upper	middle
height	i-ə	1.00000	1.00000	1.0000
	y-ə	1.00000	0.91688	ND
	u-ɑ	0.37080	0.03259	1.0000
	u-ɔ	1.00000	0.15022	1.0000
	i-u	0.89866	1.00000	1.0000
	y-u	0.05915	1.0000	ND
	ε-ɑ	3.8e-06	1.00000	9e-06
	ε-ɔ	0.02417	0.04722	1.0000
	ø-ɑ	0.00041	0.25790	ND
	i-y	0.00041	0.25790	ND
	ε-ø	0.00060	0.26185	0.5554
	ɑ-ɔ	1.00000	1.00000	1.0000
	u-u	0.93427	1.00000	1.0000

Turkish

V height: TB↑
V backness: TB↔
(Washington, 2016, 2019)

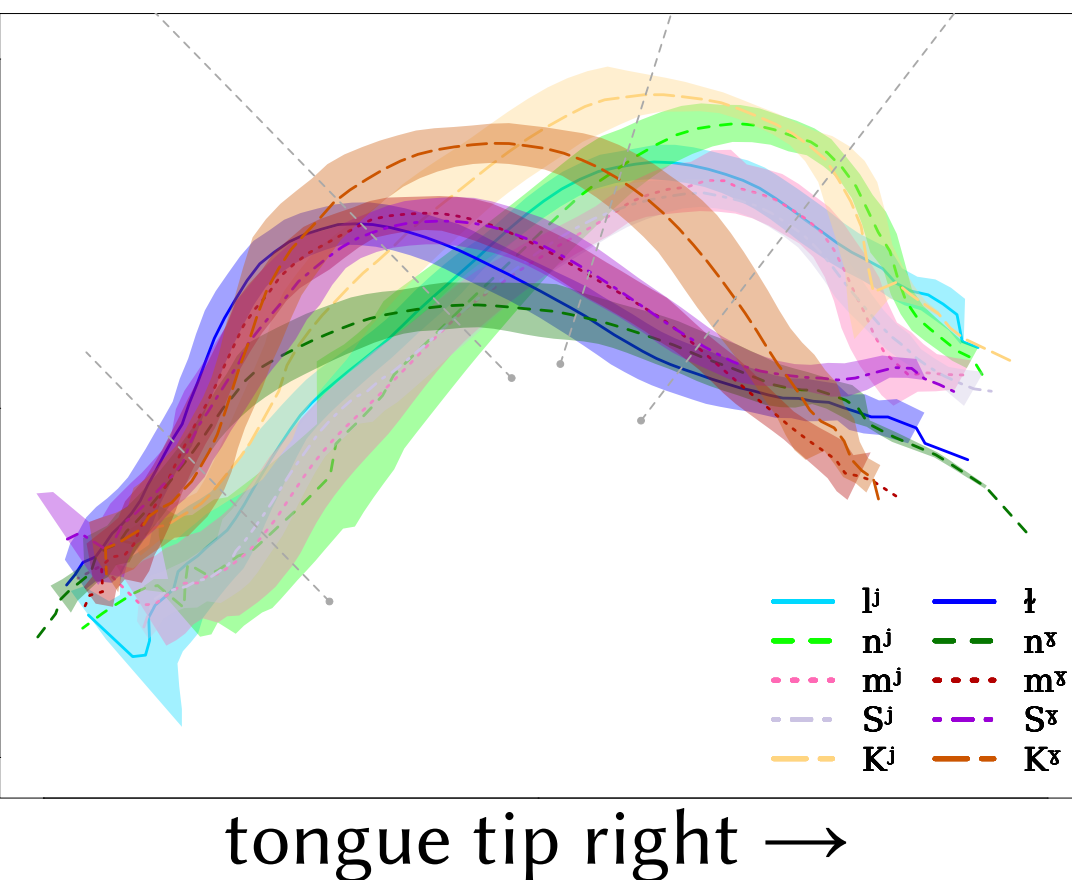


	pair	middle	upper	middle
height	i-ə	0.46115	0.0182	1.2e-07
	y-ə	0.03278	0.2266	0.00128
	u-ɑ	6.2e-06	0.7333	5.3e-09
	u-ɔ	1.8e-08	1.0000	0.00067
	i-u	4.0e-09	5.6e-07	0.00968
	y-u	1.5e-15	9.7e-15	4.1e-06
	ε-ɑ	< 2e-16	2.4e-14	< 2e-16
	ε-ɔ	2.7e-07	3.0e-13	1.5e-05
	i-y	0.07060	0.0035	0.00055
	ε-ø	0.02076	0.0013	0.02297
	ɑ-ɔ	6.0e-05	2.9e-07	0.00026
	u-u	0.00136	1.0000	0.01488

V height: TB↑, TR↔
V backness: TB↓, TR↔
V roundedness: TB↑, TR↔

Russian

C^y/C^j: TB↓, TR↔
(Banach et al., 2026, etc.)



	pair	middle	back	upper	lower
palatalisation	M	6.8e-10	0.05028	7.5e-09	0.00022
	L	<2e-16	<2e-16	<2.3e-16	2.8e-08
	N	7.2e-12	3.4e-05	1.00000	0.00648
	S	<2e-16	0.01166	5.8e-14	3.0e-07
	K	1.1e-10	0.00678	0.20871	1.00000

C^y/C^j: TB↓, TR↔ (not /K/)

Methodology

- Data** from previous studies (cited)
 - Wordlist tasks, in carrier phrases
- Annotated** midsagittal tongue surface using UltraTrace (Murphy et al., 2020)
 - frame closest to midpoint of Vs
- Plotted** and **analysed** using ultrapolaRplot (Outkin & Washington, 2024)
 - Perpendicular rays intersect various areas of the tongue (mid TB, upper TR, etc.)
 - Pairwise t-test between categories at intersections (adjusted for Type I error)
 - Pairwise t-test compares maximum curvature (PMC) groups by calculating differences in distance to origin
 - PMC comparison is a measure of difference in TB position

Discussion

- TR↔ significant factor in Russian C^y/C^j; does not hold for all pairs
- TR↔ significant factor in all Turkic V backness, but not for each individual pair in each language
- Some areas of TR significant for given contrast, while other areas not
- TR↔ significant for some rounding pairs, likely used to amplify perceptual distinctiveness beyond lip rounding

Future work

- More robust datasets
- Multiple speakers of the same language
- Extent of co-production vs TR independence
- Different analytical methods?

References

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